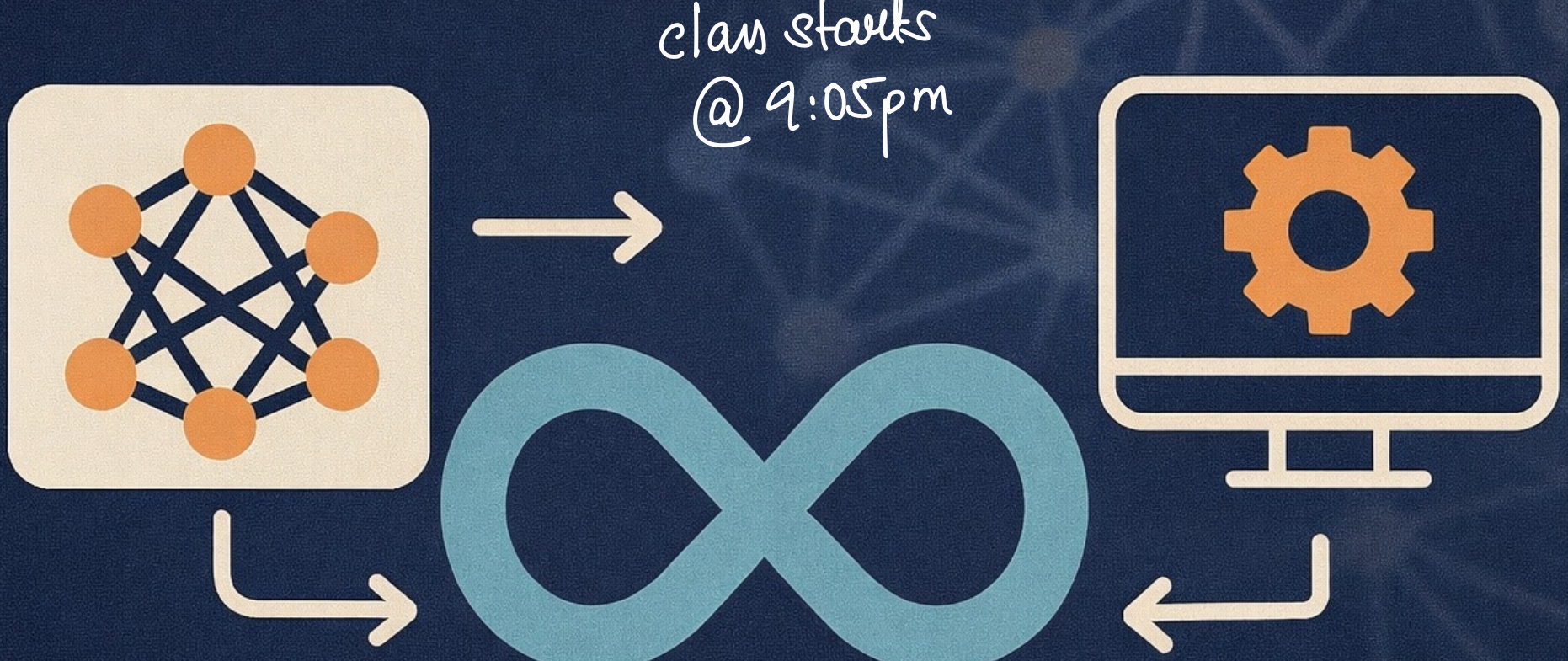
NEURAL NETWORK

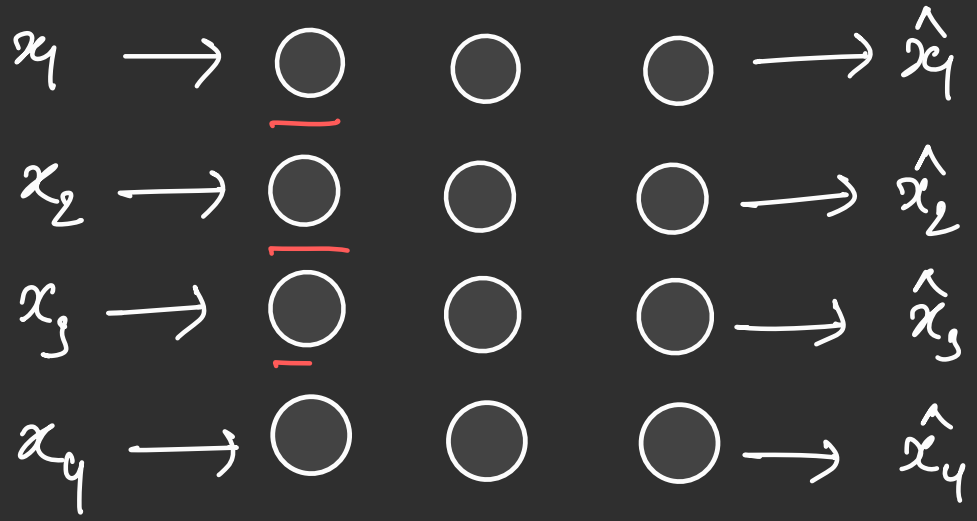


Lecture : Auto Encoder

#Agenda

1) Auto Encoders

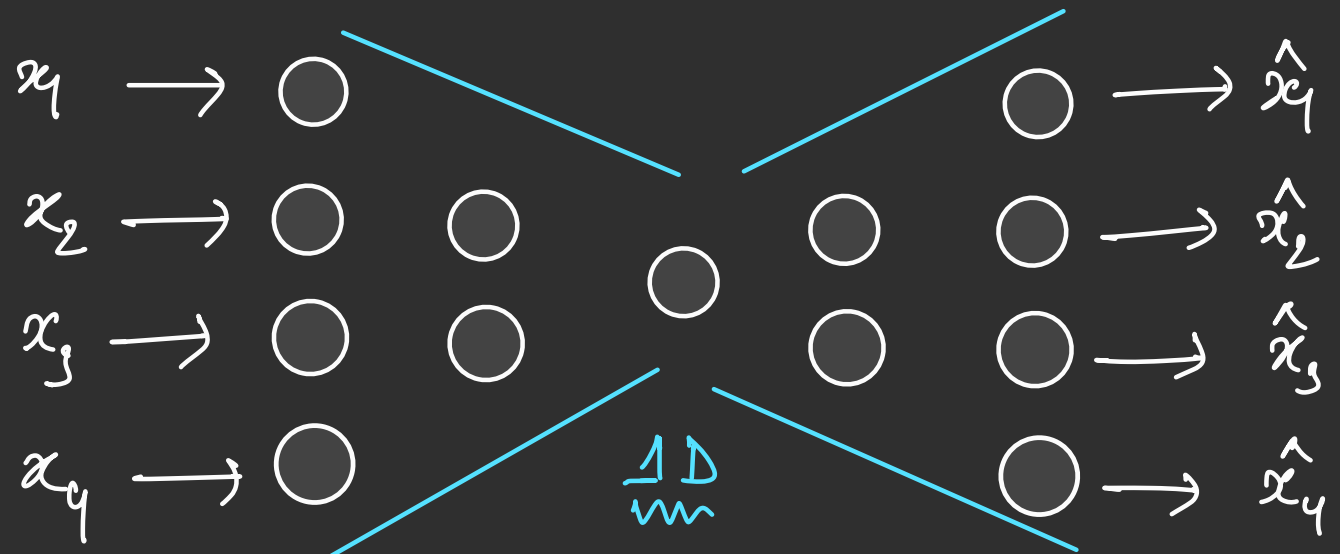
The idea



$x_i^o = \hat{x}_i^o$

↳ prediction of input

Obj ⇒ Reduce dim without much loss of info



4 dim

Encoder

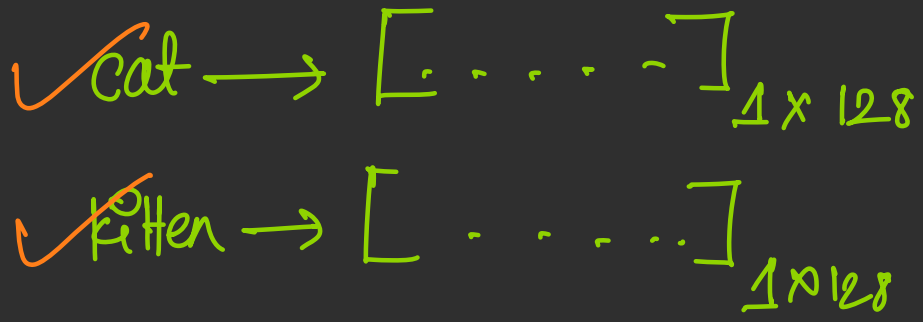
Decoder

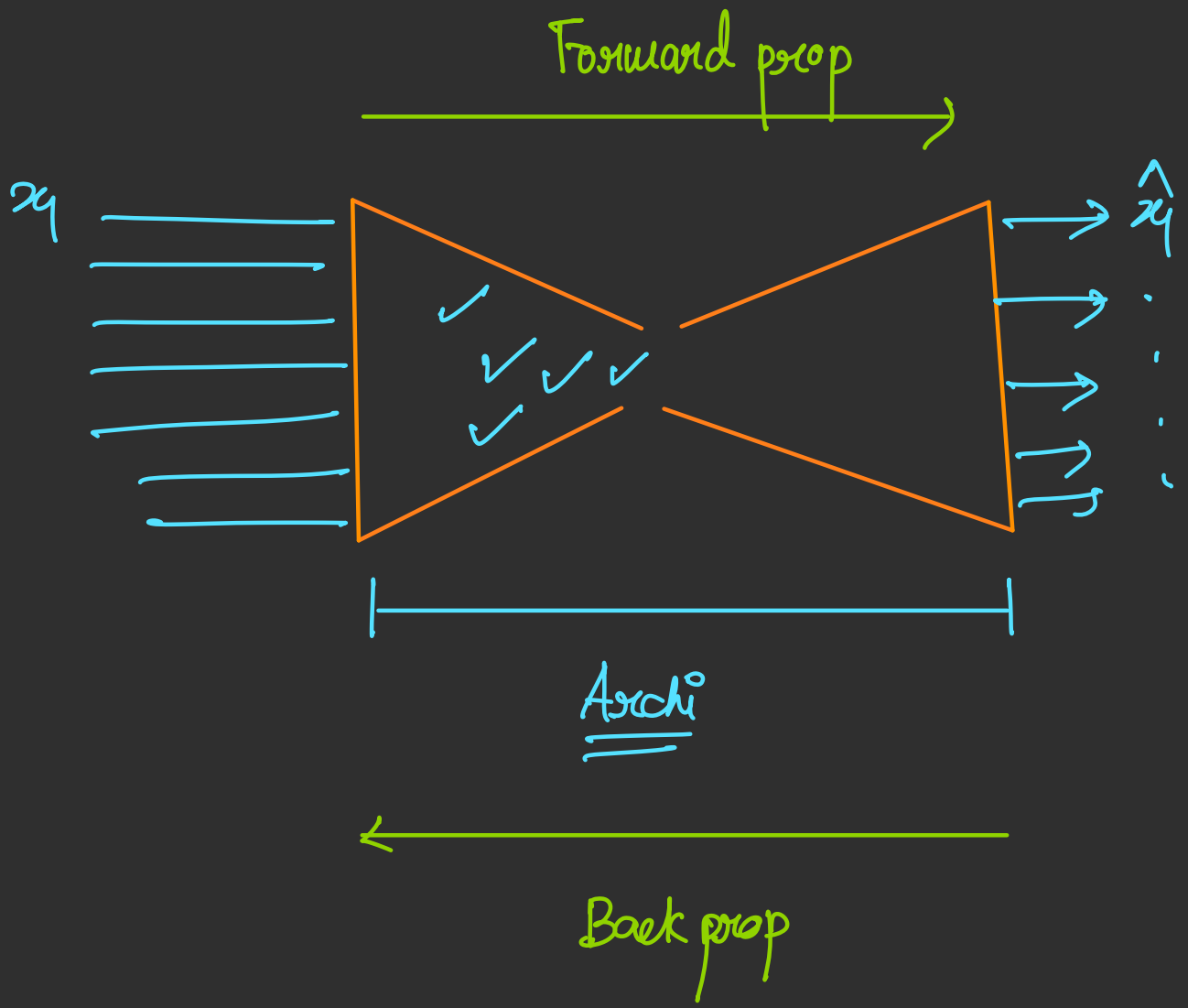
"unzip"

.zip

Use Cases

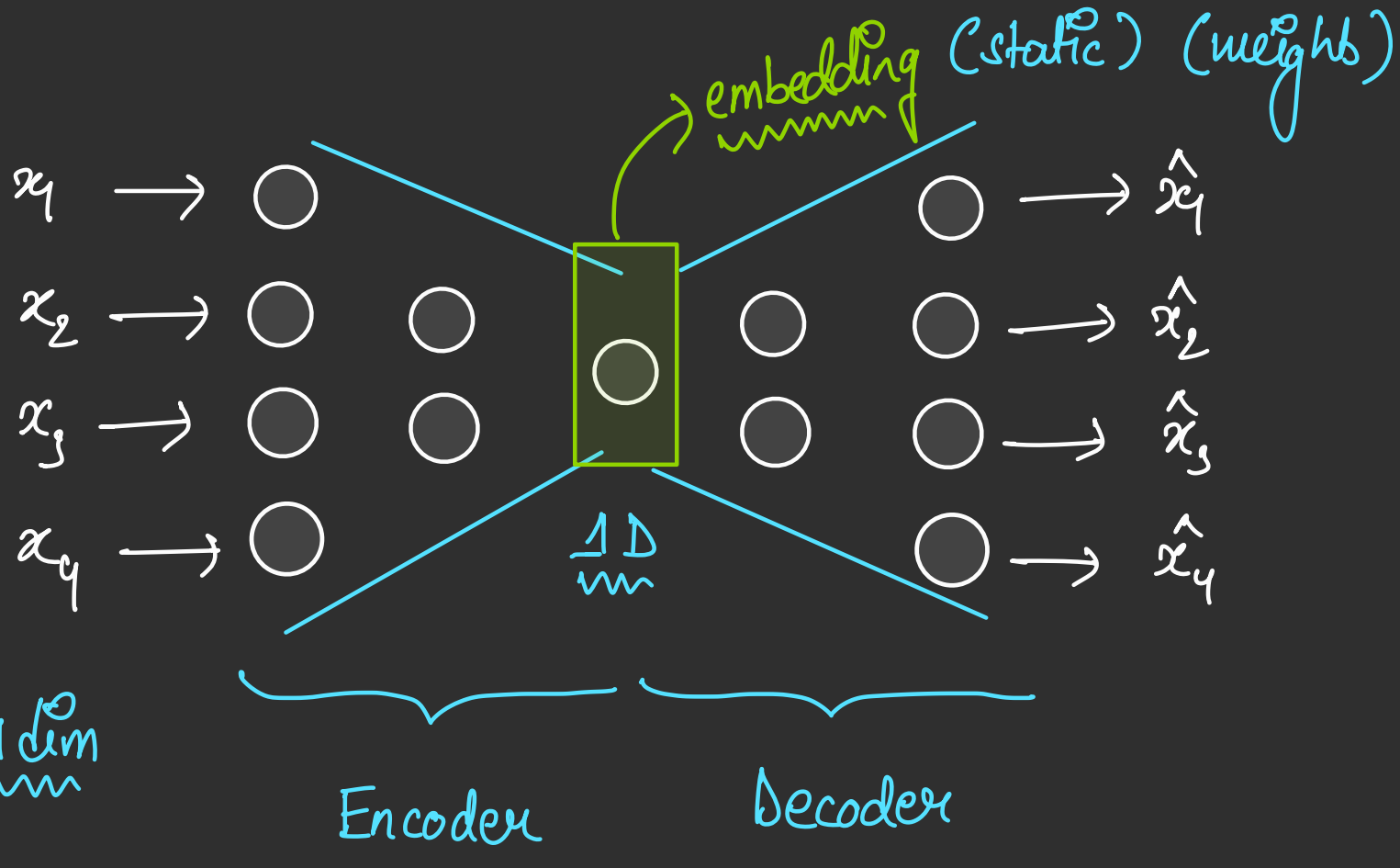
1) Compression / Dim reduction → faster inference time & low latency.
2) Visualization
3) De-noising
4) Creating Embedding
    1) Recommendation Sys
    2) Clustering
    3) NLP — Information Retrieval (System Search)


cat → [. . . . -]
1 x 128
kitten → [. . . . .]
1 x 128



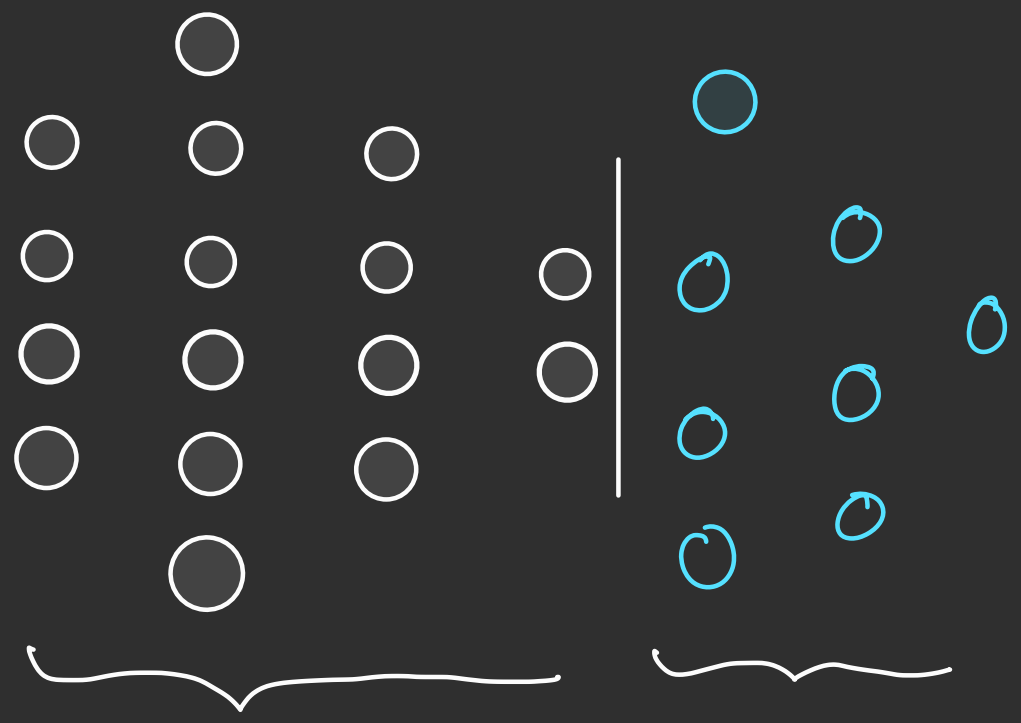
$x \neq \hat{x}$

$L \uparrow$


embedding (static) (weights)
x_1
x_2
x_3
x_4
$\hat{x}_1$
$\hat{x}_2$
$\hat{x}_3$
$\hat{x}_4$
1D
4 dim
Encoder
Decoder

AE

Transfer Learning



R- Lab → 100 M USD

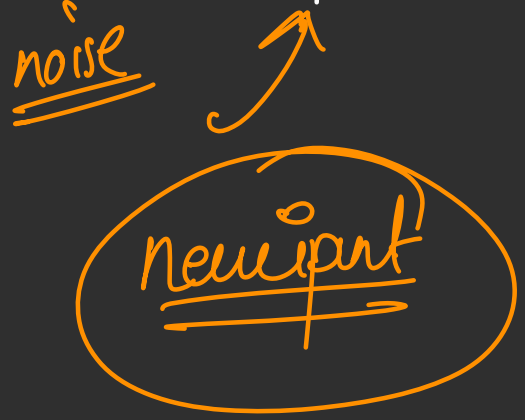
Train a model on Covid

(ouoft) input = output (same input) [Training]

input ≈ output (similar input)

noise

newinput



Neural N/w

N/w

↳ Supervised
(data + label)

Unsupervised

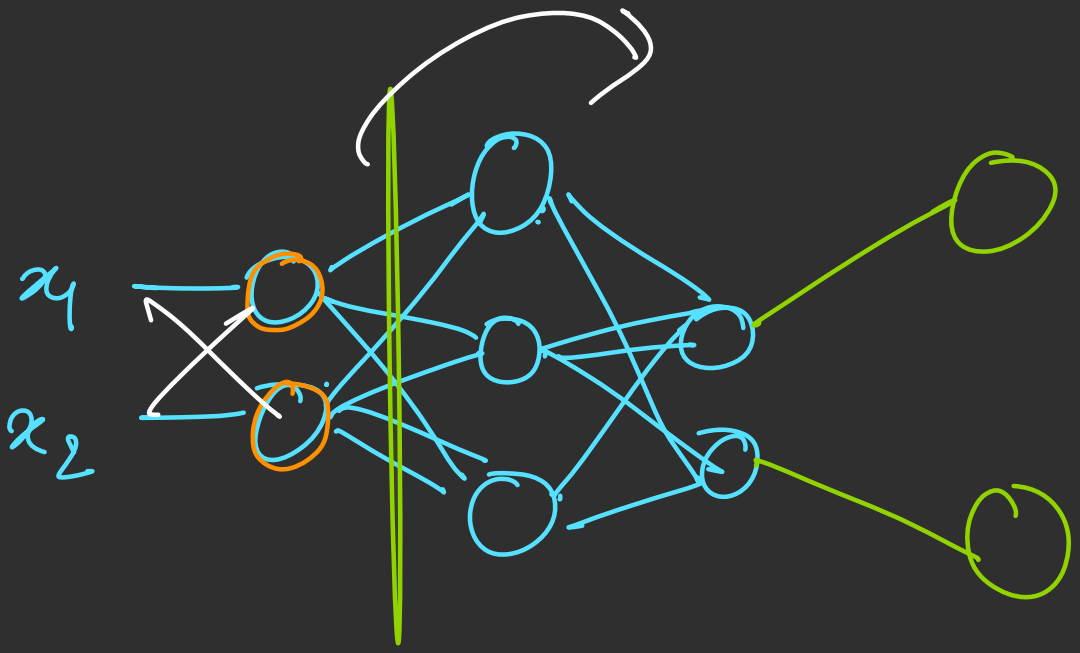
↳ Clustering, AE

Self Supervised

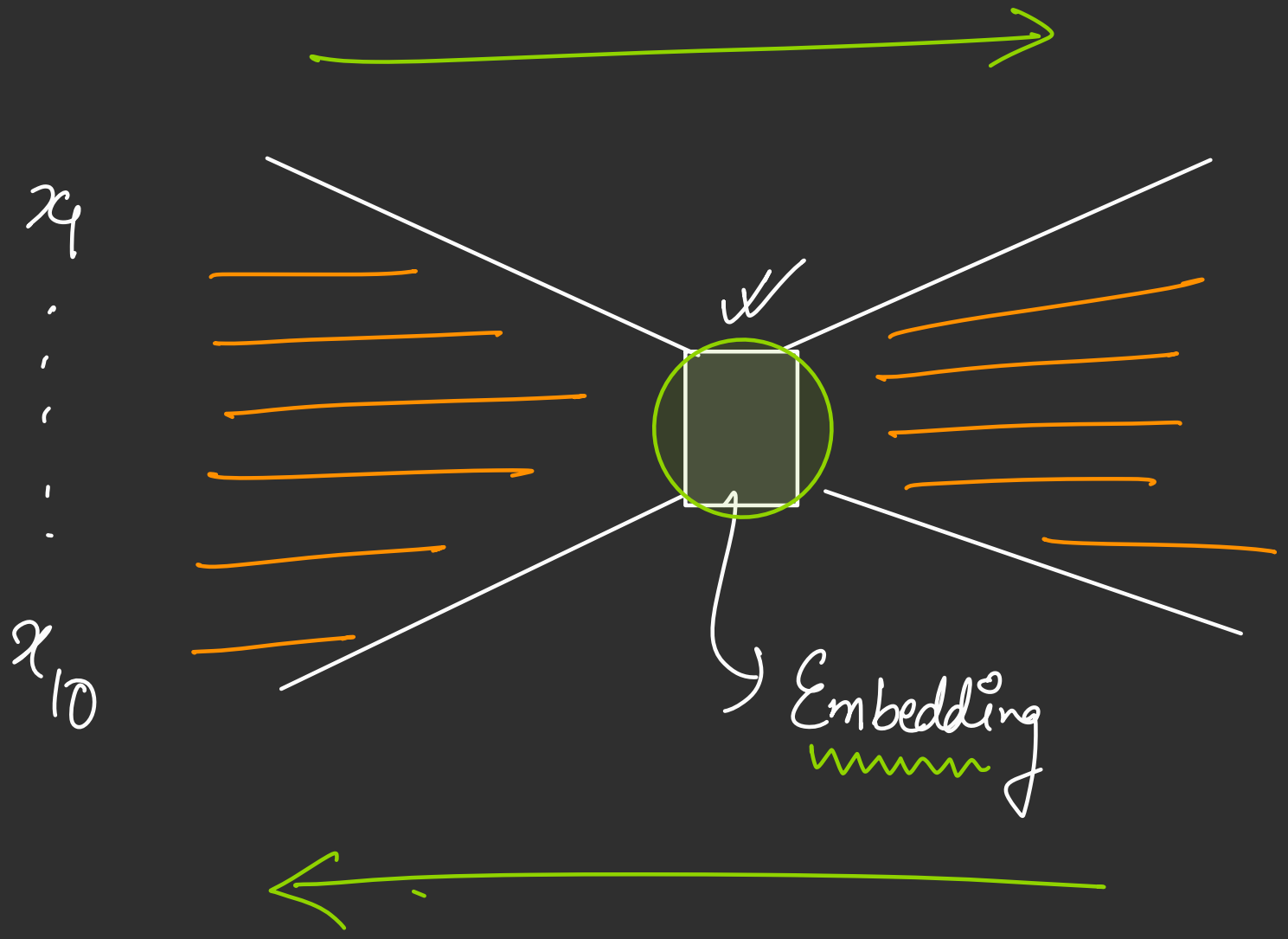
↳ labels are learnt from data itself

RL

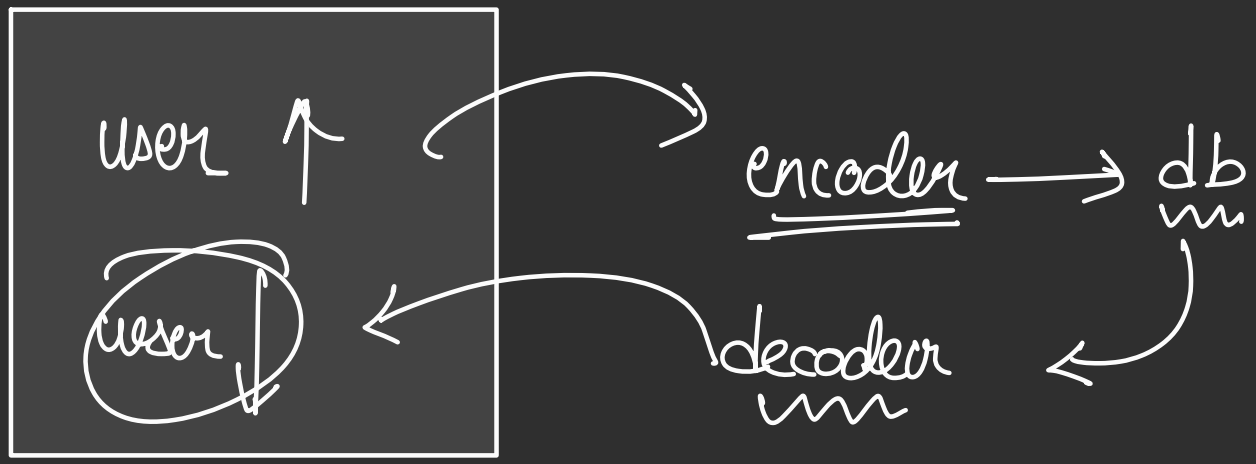
↳ N/w

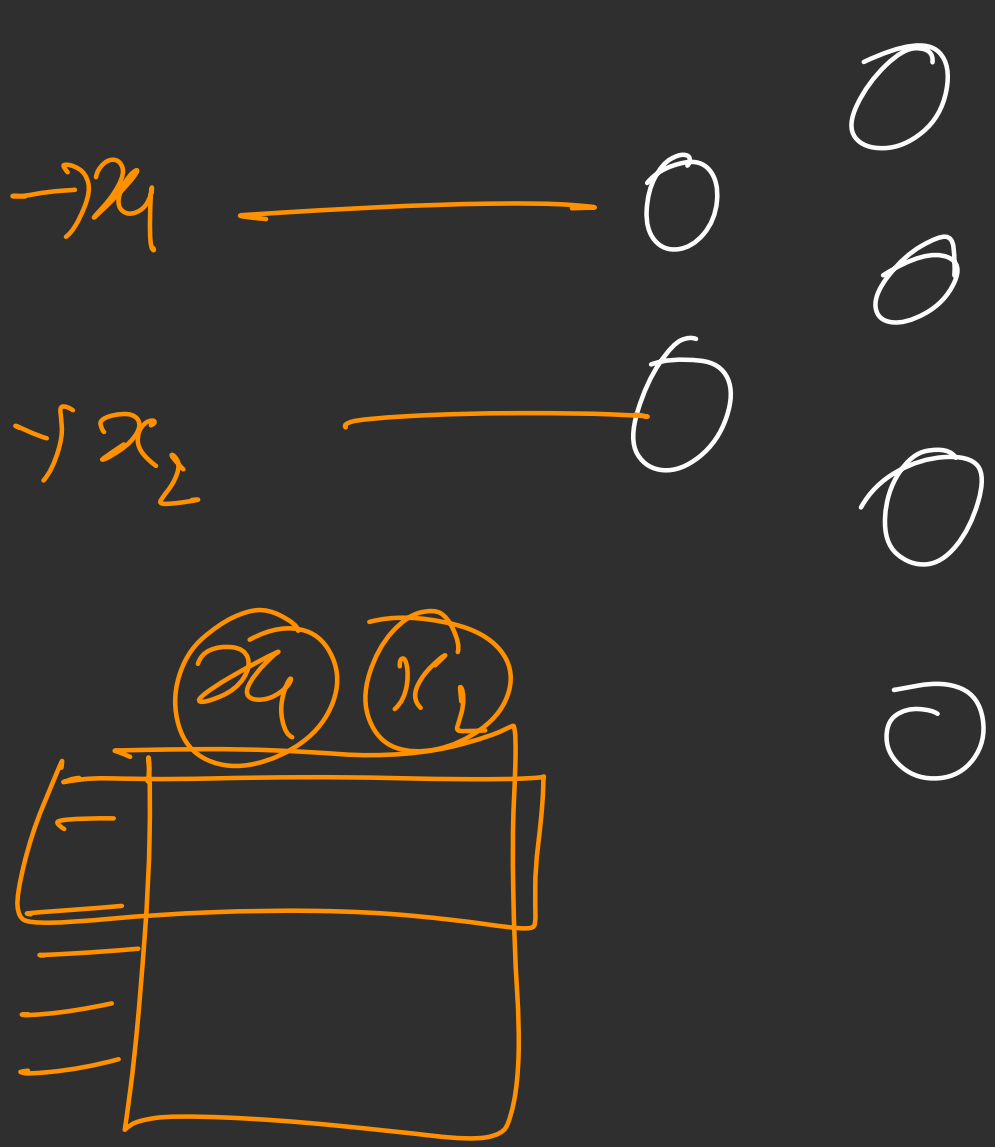


$$\sigma\left(x_1 w_1 + x_2 w_2 + b\right)$$



10 { x_1 ⋮ x_{10} } → Emb (3)


user
user
encoder
db
decoder



ANN

batch = 32

1020 rows



1020 x 64

6000

(6000 x 64)